

IHF Online Education Weeks 2023



Injury Prevention & Importance of long-term athletic plans with transfer periods between beach/indoor

By Alexander Novakovic

Head coach German women's national team

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Alexander Novakovic

**1984 in Munich / Germany*

Married / 2 sons / air traffic controller

2015 - 2019 coach of German youth national teams (4 medals)

Since 2018 Head coach German women's national team

European Champion 2021

World Champion 2022 & World Games Champion 2022

Agenda



01 *Injury Overview*
What Happened Most?

02 *Long-Term Athletic Cycle*
1-Year Term

03 *Regeneration -*
Remember all Aspects

04 *Transfer Periods Between*
Beach/Indoor

05 *Questions*



01

Injury Overview



What Happened Most?

Data from GER national teams 2019-2022

- *lower injury rates than in indoor handball*
- *no injury with more than 4 weeks out (EURO 2019 & 2021)*
- *overuse injuries (esp. after transfer)*
- higher energy costs for training on sand



What can coaches do?

Task: Nearly kill all injury aspects

1

Individual long-term athletic plans considering beach/indoor load and requirements → **preparation**

2

Individual transition period between surfaces for each player → **protection**

3

High focus on individual load control, esp. on the players who play indoor/beach at the same time → **regeneration**



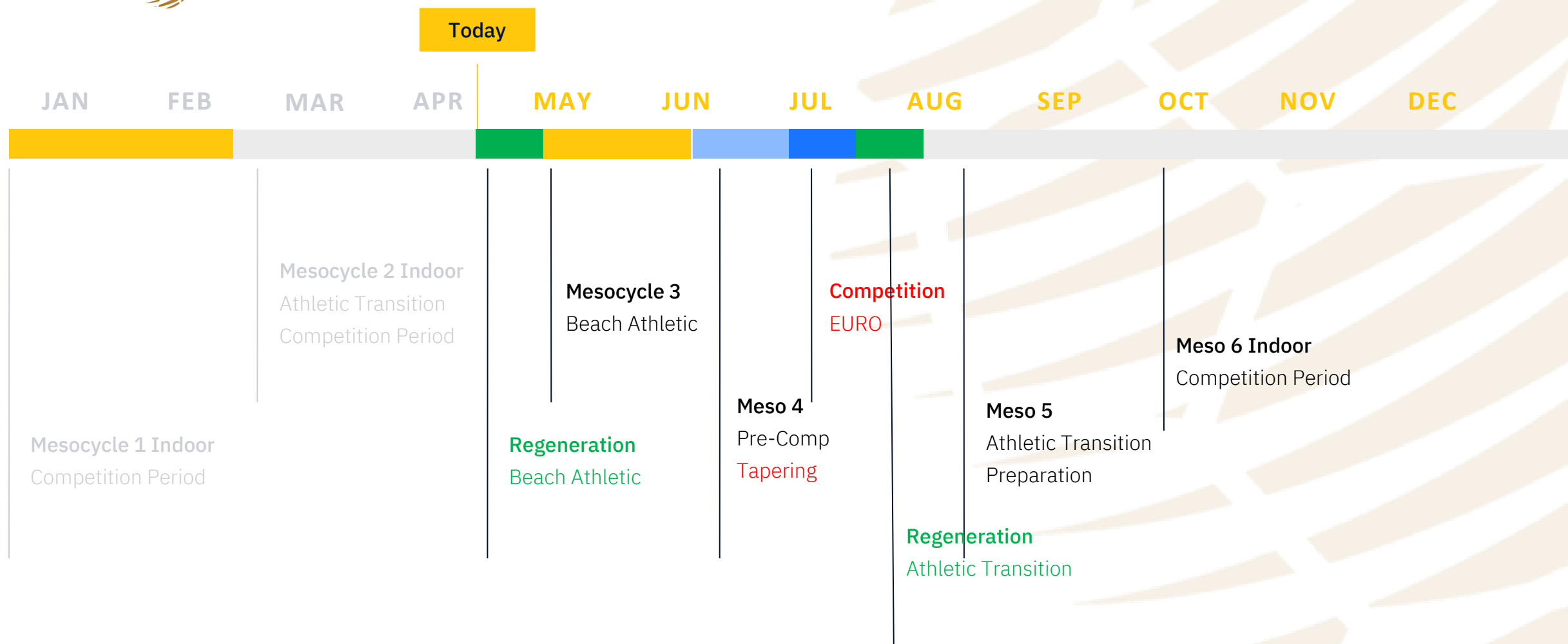


02

Long-Term Athletic Cycle



Create a long-term cycle with indoor/beach requirements





How?

TOGETHER

Build an athletic plan
or adjust an existing plan

- Indoor & Beach requirements (specific)
- Whole season together with all coaches who take care of your athlete.

ADVANTAGES FOR ALL

Change of surface in the last indoor period leads to a performance plus e.g. in jumping and speed starting.

Why?

On sand, different and partly new muscle parts are addressed.



AT THE END

Higher trust between coaches

Less prejudices & better communication

Better knowledge of indoor coaches about beach

Better prevention & less injuries



Better athlete on all surfaces



Communication

Important issues

During indoor season:

- Close communication is obligatory, maybe visit indoor club once a year
- If an athlete has to play a lot (due to other injuries in team) discuss with club whether a nomination makes sense.
- When in doubt give the athlete a break.
- You can get the athlete's opinion, but from my experience the athlete always wants to play.



Creates Trust



03

Regeneration



100 points per day!

More regeneration = better performance + less injuries = success



<i>Category: sleep</i>	<i>points</i>
9 hours of sleep per night (-10 points for every hour less)	20
30min powernap at noon	10
30min before sleep, no mobile, TV, computer	10
<i>Category: nutrition</i>	
Breakfast high in protein (eggs, dairy, e.g.)	10
3 litres of water (without training), min. 50-200mg/l sodium	10
3 servings of fruits and vegetables	10
Post-workout shake	10
25g protein before going to sleep	10
No alcohol, industrial sugar, white flour	10
<i>Category: active & passive recreation</i>	
Contrast shower (hot-cold-hot-cold); 4-6 times for 30 sec.	10
Physiotherapy / massage	15
Active regeneration (running, swimming, cycling) min. 30min	15
Mobility exercises, stretching, light weight training	10



Nutrition - most important facts

Vitamin D

2000 IE daily

- mood!
- muscle function
- defence

Vitamin C

500 mg daily

- protection against stress
- immune defence
- regeneration

Omega 3

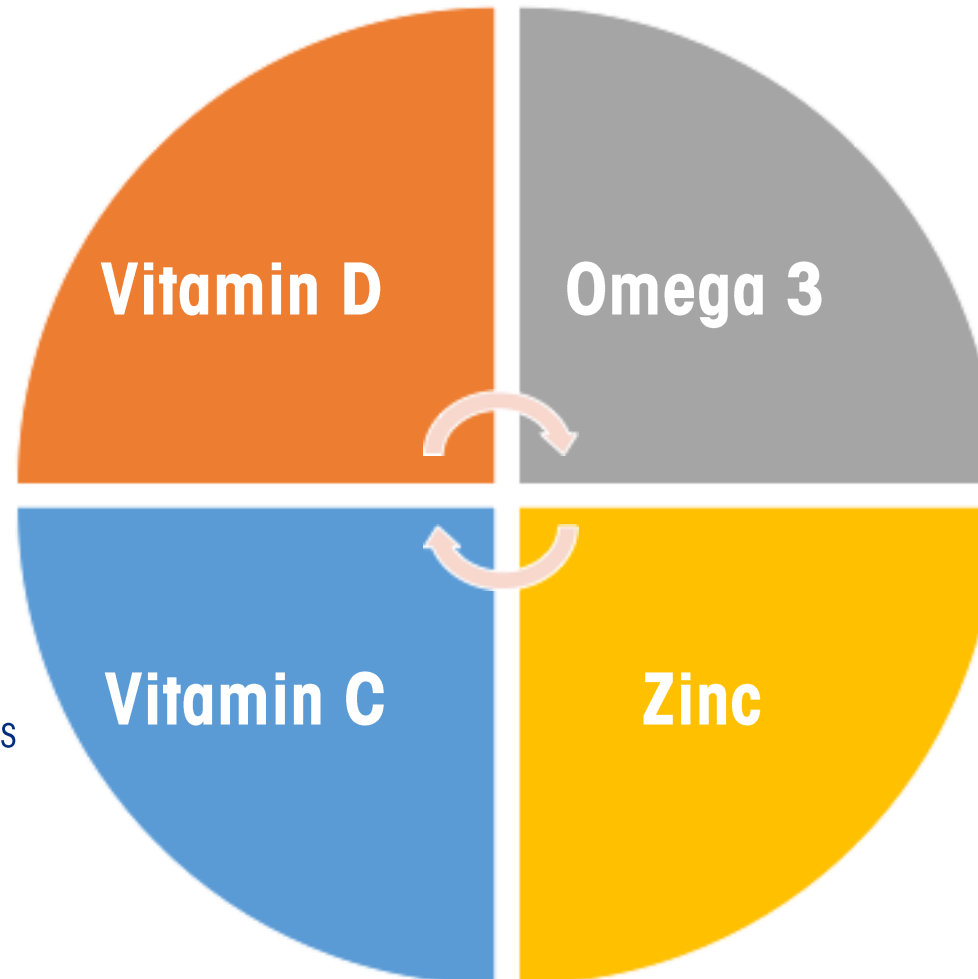
1000mg daily

- cell protection
- regeneration

Zinc

10mg daily

- hormone formation
- immune defence





04

Transfer Periods Between Beach/Indoor



What can coaches do?

Task: Nearly kill all injury aspects

1

Individual long-term athletic plans considering beach/indoor load and requirements —> **preparation**

Individual transition period between surfaces for each player —> **protection**

3

High focus on individual load control, esp. on the players who play indoor/beach at the same time —> **regeneration**





Transition period

Create it!

- between seasons/phases of different surfaces
- between indoor/beach seasons, a period of 6 weeks should be created (best outcome)
- adaption possible

No need to break, include it!

No need to train extra, adapt it!

PLYOMETRICS

SPRINTS / HITs / CHANGE-OF-DIRECTION RUN

ROTATION WORK





NPS SURVEY 2022

... and back

- prepare load on indoor surface during last beach weeks
- prepare 1 against 1 with body contact
- min. 5 days of transition break is recommended
- include transition in weekly plans



1. Woche	2. Woche	3. Woche	4. Woche	5. Woche	6. Woche
Sand					
Halle					



Short transition

During indoor season after beach camp

To sand

Prepare

- Rotation movements
- Higher number of jumps

Be attentive

- Overload in rotation movements (back)
- Overload in jumping movements (legs, feet)

To indoor

Prepare full body contact

- First day of indoor training without body contact

Be communicative

- Have close contact to indoor coaches
- Establish close contact between beach physio and indoor physio

ABC Running Drills



Start & Stop



Plyometrics 1



Plyometrics 2





05

Questions



„We have to educate coaches that the typical narratives like 'beach handball leads to injuries' or 'beach handball is bad for your indoor career' are wrong.

With good communication and load control, every beach athlete will benefit for indoor handball! In athletic, technical, tactical and decision-making issues!“

Alexander Novakovic



Thank you.

Contact:

alexander.novakovic@dhb.de



Instagram: alex10nova